$$= \sqrt{\frac{\lim_{x\to\infty} 3 + \lim_{x\to\infty} \frac{1}{x^3}}{1 - 8 \lim_{x\to\infty} \frac{1}{x^2}}}$$

$$= \frac{\sqrt{3+0}}{1-0} = \sqrt{3}$$

Answer

$$\lim_{x \to -\infty} \frac{\sqrt{\frac{3x^4}{x^4} + \frac{x}{x^3}}}{\frac{x^2 - 8}{x^2}}$$

$$= \lim_{x \to -\infty} \frac{\sqrt{3 + \frac{1}{x^3}}}{1 - \frac{8}{x^2}}$$

$$= \frac{\sqrt{\lim_{x \to -\infty} 3 + \lim_{x \to -\infty} \frac{1}{x^3}}}{1 - 8 \lim_{x \to -\infty} \frac{1}{x^2}}$$

$$= \frac{\sqrt{3+0}}{1-0} = \sqrt{3}$$

Question 30

$$\lim \frac{\sqrt{3x^4 + x}}{x^2 - 8}$$

Divide by x^2

$$= \lim_{x \to \infty} \frac{\sqrt{3 + \frac{1}{x^3}}}{1 - \frac{8}{}}$$

$$\frac{0-(-1)}{\sqrt{0+6}} = \frac{1}{\sqrt{6}}$$ ans

Question 28

$$\lim_{y\to\infty} \frac{2-y}{\sqrt{7+6y^2}}$$

Divide by y^2

$$\lim \frac{\frac{2}{y} - y}{\sqrt{\frac{7}{y^2}+6}}$$

$$2 \lim_{y\to\infty} y - y$$

$$7 \lim_{y\to\infty} y^2 + 6$$

$$= \frac{0-1}{\sqrt{0+6}} = \frac{-1}{\sqrt{6}}$$

Answer

Qno 29 $\lim_{x\to\infty} \frac{\sqrt{3x^4+x}}{x^2-8}$

$$= \sqrt{\frac{\lim_{x\to\infty}\left(5-\frac{2}{x^2}\right)}{\lim_{x\to+\infty}\left(\frac{1}{x}+\frac{3}{x^2}\right)}}$$

$$= \sqrt{\frac{5-2\lim_{x\to\infty}\frac{1}{x^2}}{\lim_{x\to+\infty}\frac{1}{x}+3\lim_{x\to\infty}\frac{1}{x^2}}}$$

$$= \frac{\sqrt{5-0}}{1+0} = \sqrt{5} \text{ Ans}$$

Question NO 27

$$\lim_{y\to-\infty}\frac{2-y}{\sqrt{7+6y^2}}$$

Divided by $|y|$

$$\lim_{y\to\infty}\frac{\frac{2}{|y|}-\frac{y}{|y|}}{\sqrt{\frac{7}{y^2}+6}}$$

since $y<0$

$$\frac{y}{|y|} = -1, \quad \frac{2}{|y|}\to 0 \quad \frac{7}{y^2}\to 0$$

Question 25

$$\lim_{x \to \infty} \frac{\sqrt{5x^2-2}}{x+3}$$

Divid by $|x|$

$$\lim_{x \to -\infty} \frac{\sqrt{5 - \frac{2}{x^2}}}{\frac{x}{|x|} + \frac{3}{|x|}}$$

$x < 0 \qquad \frac{x}{|x|} = -1$

$\frac{2}{x^2} \to 0 \quad , \quad \frac{3}{|x|} \to 0$

$$= \frac{\sqrt{5-0}}{-1+0} = -\sqrt{5}$$

26) $\lim_{x \to +\infty} \frac{\sqrt{5x^2-2}}{x+3}$

$$= \lim_{x \to \infty} \frac{\sqrt{\frac{5x^2}{x^2} - \frac{2}{x^2}}}{\frac{x}{x^2} + \frac{3}{x^2}}$$

$$\lim \frac{1}{x^2} + \lim 8$$

$$= \sqrt{\frac{2(0) + 3(0) - 5}{8}}$$

$$= \sqrt[3]{\frac{-5}{8}} = -3\sqrt{\frac{5}{8}}$$ Ans

Question NO 24:-

$$\lim_{s \to \infty} \sqrt[3]{\frac{3s^7 - 4s^5}{2s^7 + 1}}$$

Divide by s^7

$$= \sqrt[3]{\frac{\lim_{s \to \infty}\left(3 - \frac{4}{s^2}\right)}{\lim_{s \to \infty}\left(2 + \frac{1}{s^7}\right)}}$$

$$= \sqrt[3]{\frac{3-0}{2+0}} = \sqrt[3]{\frac{3}{2}}$$ Answer

Divide by x^3

$$\lim_{x \to -\infty} \frac{\frac{1}{x^2} + 4}{\frac{1}{x^3} - \frac{1}{x} + 7}$$

$$= \frac{\lim_{x \to \infty} \frac{1}{x^2} + \lim_{x \to \infty} 4}{\lim_{x \to \infty} \frac{1}{x^2} - \lim_{x \to \infty} \frac{1}{x} + 7}$$

$$= \frac{0 + 4}{0 - 0 + 7} = \frac{4}{7}$$

Question NO 23 :-

$$\lim_{x \to \infty} \sqrt[3]{\frac{2 + 3x - 5x^2}{1 + 8x^2}}$$

Divide by x^2

$$= \sqrt[3]{\lim_{x \to +\infty} \frac{\frac{2}{x^2} + \frac{3}{x} - 5}{\frac{1}{x^2} + 8}}$$

$$= \sqrt[3]{\frac{2 \lim_{x \to \infty} \frac{1}{x^2} + 3 \lim_{x \to \infty} \frac{1}{x} - 5}{}}$$

$= \frac{0+\infty}{0} = +\infty$ Ans

Question no 21

$$\lim_{t \to +\infty} \frac{6 - t^3}{7t^3 + 3}$$

Divide by t^3

$$= \lim_{t \to +\infty} \frac{\frac{6}{t^3} - 1}{7 + \frac{3}{t^3}}$$

$$= \frac{6 \lim_{t \to \infty} t^3 - 1}{7 + 3 \lim \frac{t}{t^3}}$$

$$= \frac{0 - 1}{7 + 0} = \frac{-1}{7}$$

Question NO 22

$$\lim_{x \to -\infty} \frac{x + 4x^3}{1 - x^2 + 7x^3}$$

$$= \frac{0 - \infty}{1}$$

$$= -\infty \text{ answer}$$

20) $\lim_{t \to -\infty} \frac{5 - 2t^3}{t^2 + 1}$

Divide by t^2

$$\lim_{t \to -\infty} \frac{\frac{5}{t^2} - \frac{2t^3}{2t^2}}{\frac{t^2}{t^2} + \frac{1}{t^2}}$$

$$= \frac{\lim_{t \to -\infty} \left(\frac{5}{t^2} - 2t\right)}{\lim_{t \to -\infty} \left(\frac{1}{t^2}\right)}$$

$$= \frac{5 \lim_{t \to -\infty} \frac{1}{t^2} - 2 \lim_{t \to -\infty}}{\lim_{t \to -\infty} \left(\frac{1}{t^2}\right)}$$

$$= \frac{5(0) - (-2)(-\infty)}{0}$$

18) $\lim_{x \to +\infty} \frac{5x^2+7}{3x^2-x}$

$= \lim_{x \to +\infty} \frac{\frac{5x^2}{x^2} + \frac{7}{x^2}}{\frac{3x^2}{x^2} - \frac{x}{x^2}}$

$= \lim_{x \to +\infty} \frac{5 + \frac{7}{x^2}}{3 - \frac{1}{x}}$

$= \frac{\lim_{x \to +\infty} 5 + 7 \lim_{x \to +\infty} \frac{1}{x^2}}{\lim_{x \to +\infty} 3 - \lim_{x \to +\infty} \frac{1}{x}}$

$= \frac{5+0}{3-0} = \frac{5}{3}$

19) $\lim_{x \to +\infty} \frac{7-6x^5}{x+3}$

Divide by x

$= \lim_{x \to +\infty} \frac{\frac{7}{x} - 6x^4}{1 + \frac{3}{x}}$

$$y + 4 \to -\infty$$

$$= \frac{3}{-\infty} = 0$$

$$\lim_{x \to +\infty} \frac{1}{x-12} = \frac{1}{+\infty} = 0$$

17) $$\lim_{x \to -\infty} \frac{x-2}{x^2+2x+1}$$

Divided by x^2

$$\lim_{x \to -\infty} \frac{\frac{1}{x} + \frac{2}{x^2}}{1 + \frac{2}{x} + \frac{1}{x^2}}$$

$$\lim_{x \to -\infty} \frac{1}{x} - 2 \lim_{x \to \infty} \frac{1}{x^2}$$

$$1 + 2 \lim_{x \to -\infty} \frac{1}{x} + \lim_{x \to -\infty} \frac{1}{x^2}$$

$$= \frac{0-0}{1+0+0} = 0$$

$$\lim_{x \to +\infty} \frac{5x^2 - 4x}{2x^2 + 3}$$

Divide by x^2

$$\lim_{x \to +\infty} \frac{\frac{5x^2}{x^2} - \frac{4x}{x^2}}{\frac{2x^2}{x^2} + \frac{3}{x^2}}$$

$$= \frac{\lim_{x \to \infty} 5 - \frac{4}{x}}{2 + \frac{3}{x^2}}$$

$$= \frac{\lim_{x \to \infty} 5 - 4 \lim_{x \to \infty} \frac{1}{x}}{\lim_{x \to \infty} 2 + 3 \lim_{x \to \infty} \frac{1}{x^2}}$$

$$= \frac{5 - 4(0)}{2 + 0} = \frac{5 - 0}{2 + 0}$$

$$= \frac{5}{2} \text{ answer}$$

$$\lim_{y \to \infty} \frac{3}{y + 4}$$

$y \to \infty$

Therefore

$-\sqrt[3]{42}$

f) $\lim_{x \to -\infty} \frac{g(x)}{f(x)}$

$$\frac{\lim_{x \to -\infty} g(x)}{\lim_{x \to -\infty} f(x)} = \frac{-6}{7}$$ answer

g) $\lim_{x \to -\infty} \left[f(x) + \frac{g(x)}{x} \right]$

$$\lim_{x \to -\infty} f(x) + \lim_{x \to -\infty} \frac{g(x)}{x}$$

$$= 7 + \frac{-6}{-\infty}$$

$$= 7 + 0 \qquad = 7$$ answer

h) $\lim_{x \to -\infty} \frac{x f(x)}{(2x+3)(g(x))}$

Divide by x

$$\lim_{x \to \infty} \frac{\frac{x}{x} f(x)}{\left(\frac{2x}{x} + \frac{3}{x}\right) g(x)}$$

$$\lim_{x \to \infty} \frac{f(x)}{2 + \frac{3}{x} + g(x)}$$

$$\frac{\lim f(x)}{2 + 3 \lim \frac{1}{x} + g(x)}$$

$$\frac{7}{2 + 3(0) + (-6)} = \frac{7}{-12} \quad \text{Answer}$$

Question NO 9:-

Find the limits:-

9) $\lim_{x \to -\infty} (1 + 2x - 3x^5)$

highest degree of polynomial

$= -3x^5$

$= -3(-\infty)^5$

$= -3(-\infty)$

$= \infty$

$$\lim_{x \to -\infty} (2x^3 - 100x^2 + 5)$$

highest degree of polynomial

$x \to -\infty$

$2x^3 \to -\infty$

$-\infty$

$$\lim_{x \to +\infty} (\sqrt{x})$$

$\sqrt{x} \to +\infty$

$= +\infty$

$$\lim_{x \to -\infty} \sqrt{5-x}$$

As $x \to -\infty$

$= \sqrt{5-(\infty)}$

$= +\infty$

3 $\lim_{x \to \infty} h(x) + 4$

$\lim_{x \to \infty} x^2$

$= \frac{3(0) + 4}{+\infty} = \frac{4}{+\infty} = 0$

h) $\lim_{x \to +\infty} \frac{6f(x)}{5f(x) + 3g(x)}$

$\frac{6 \lim_{x \to +\infty} f(x)}{5 \lim_{x \to \infty} f(x) + 3 \lim_{x \to \infty} g(x)}$

$= \frac{6(3)}{5(3) + 3(-5)} = \frac{18}{15 - 15}$

$= \frac{18}{0} = \infty$

Question NO 6 :-

Given $\lim f(x) = 7$

$\lim_{x \to \infty} g(x) = -6$

a) $\lim_{x \to \infty} [2f(x) - g(x)]$

$2 \lim_{x \to \infty} f(x) - \lim_{x \to \infty} g(x)$

$= 2(7) - (-6)$

$= 14 + 6$

$= 22$

b) $\lim_{x \to -\infty} [6f(x) + 7g(x)]$

$6 \lim_{x \to -\infty} f(x) + 7 \lim_{x \to -\infty} g(x)$

$= 6(7) + 7(-6)$

$= 42 - 42$

$= 0$

c) $\lim_{x \to -\infty} [x^2 + g(x)]$

$\lim_{x \to -\infty} x^2 = +\infty$

and $\lim_{x \to -\infty} g(x) = -6$

So, $\infty \times -6$

$= -\infty$

d) $\lim_{x \to -\infty} [x^2 g(x)]$

using product law

$= \left(\lim_{x \to -\infty} x^2\right)\left(\lim_{x \to -\infty} g(x)\right)$

$= (+\infty)(-6)$

$= -\infty$

e) $\lim_{x \to -\infty} \sqrt[3]{f(x)g(x)}$

using product law

$= \sqrt[3]{\left(\lim_{x \to -\infty} f(x)\right)\left(\lim_{x \to -\infty} g(x)\right)}$

$= \sqrt[3]{(7)(-6)} = \sqrt[3]{-42}$

d) $\lim_{x \to +\infty} [g(x)]^2$

$\left[\lim_{x \to +\infty} g(x)\right]^2$

$= (-5)^2 = 25$

e) $\lim_{x \to +\infty} \sqrt[3]{5 + f(x)}$

$\sqrt[3]{5 + \lim_{x \to +\infty} f(x)}$

$= \sqrt[3]{5 + 3}$

$= \sqrt[3]{8} = 2$

f) $\lim_{x \to +\infty} \frac{3}{g(x)}$

$\frac{\lim_{x \to \infty} 3}{\lim_{x \to \infty} g(x)} = \frac{3}{-5}$ Answer

g) $\lim_{x \to +\infty} \frac{3h(x) + 4}{x^2}$

نمبر: تاریخ: / /

a) $\lim_{x \to +\infty} [f(x) + 3g(x)]$

$\lim_{x \to +\infty} f(x) + 3 \lim_{x \to +\infty} g(x)$

$= 3 + 3(-5)$

$= 3 - 15$

$= -12$

b) $\lim_{x \to +\infty} [h(x) - 4g(x)]$

$\lim_{x \to +\infty} h(x) - 4 \lim_{x \to +\infty} g(x)$

$= 0 - 4(-5)$

$= 20$ answer.

c) $\lim_{x \to +\infty} [f(x) g(x)]$

$\lim_{x \to +\infty} f(x) \cdot \lim_{x \to +\infty} g(x)$

$= 3 \times (-5)$

$= -15$

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$= 5 \lim \frac{1}{x^2} - 2 \lim x + \lim \frac{1}{x}$

$\lim \frac{1}{x} - \lim 3$

$= \frac{5(0) - 2(0) + 0}{0 - 3} = \frac{0}{-3}$

$= -\infty$ answer

Exercise set 1.3

Question NO. 5

Given that :-

$\lim_{x \to +\infty} f(x) = 3 \quad \lim_{x \to +\infty} g(x) = -5$

$\lim_{x \to +\infty} h(x) = 0$

Find the limits that exist, if the limit does not exist explain why;

solution:

$$\lim_{x \to +\infty} 2x^5 = 2(\infty)^5 = \infty$$

$$\lim_{x \to -\infty} 2x^5 = 2(-\infty)^5 = -\infty$$

Example no 6 :-

$$\lim_{x \to -\infty} (7x^5 - 4x^3 + 2x - 9)$$

$$= \lim_{x \to -\infty} 7x^5 = -\infty$$

$$\lim_{x \to -\infty} (-4x^8 + 17x^3 - 5x + 1)$$

$$\lim_{x \to -\infty} (-4x^8)$$

$$= -4(-\infty)^8$$

$$= -\infty$$

Example no 7 :-

$$\lim_{x \to +\infty} \frac{3x + 5}{6x - 8}$$

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Divided by x

$$\lim_{x \to +\infty} = \frac{\frac{3x}{x} + \frac{5}{x}}{\frac{6x}{x} - \frac{8}{x}}$$

$$\lim_{x \to +\infty} \left(3 + \frac{5}{x}\right)$$

$$\lim_{x \to +\infty} \left(6 - \frac{8}{x}\right)$$

$$\lim_{x \to +\infty} 3 + 5 \lim_{x \to +\infty} \frac{1}{x}$$

$$\lim_{x \to +\infty} 6 + 8 \lim_{x \to +\infty} \frac{1}{x}$$

$$= \frac{3 + 0}{6 + 0} = \frac{1}{2} \text{ answer}$$

Example 8

$$\lim_{x \to \infty} \frac{4x^2 - x}{2x^3 - 5}$$

Divide by x^3

$$\lim_{x \to \infty} \frac{\frac{4x^2}{x^3} - \frac{x}{x^3}}{\frac{2x^3}{x^3} - \frac{5}{x^3}}$$

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$$\lim_{x \to \infty} \left(\frac{\frac{4}{x} - \frac{1}{x^2}}{2 - \frac{5}{x^3}} \right)$$

$$\frac{4 \lim_{x \to \infty} \frac{1}{x} - \lim_{x \to \infty} \frac{1}{x^2}}{\lim_{x \to \infty} 2 - 5 \lim_{x \to \infty} \frac{1}{x^3}}$$

$$\frac{0 - 0}{2 - 0} = 0$$

Example no 9 :-

$$\lim_{x \to +\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x}$$

$$\lim_{x \to +\infty} \frac{\frac{5x^3}{x} - \frac{2x^2}{x} + \frac{1}{x}}{\frac{1}{x} - \frac{3x}{x}}$$

$$\lim_{x \to +\infty} \frac{5x^2 - 2x + \frac{1}{x}}{\frac{1}{x} - 3}$$